

Lightweight Authentication System Based on Zero Knowledge Proof for IoT Devices

Ashish Umesh Khandait

Dept. of Computer Science and Engineering
COEP Technological University Pune, Maharashtra, India.
Pune, India.
khandaitau22.comp@coeptech.ac.in

Tanuja R. Pattanshetti

Dept. of Computer Science and Engineering
COEP Technological University Pune, Maharashtra, India.
Pune, India.
trp.comp@coeptech.ac.in

Abstract—In the rapidly expanding field of the Internet of Things, ensuring secure and efficient communication between IoT devices and servers is crucial. This paper henceforth designs a lightweight authentication system under the Zero Knowledge Proof (ZKP) protocol, utilizing the HMAC hashing algorithm for the improvement of security measures put in place. This system is designed to enable the authentication of IoT devices without sending their actual password, thus avoiding risks due to password interception at the point of authentication. For instance, if using the phrase "God is great" as the password, one can get it hashed into HMAC with this method and still be assured that it is secure even if proof is intercepted. Simulations in the simulator shall evaluate the system's effectiveness and performance. Performance metrics include accuracy, response time, memory consumption, latency, and throughput. The presented results intend to serve lightweight device authentication mechanisms for IoT devices such that the trustworthiness of IoT devices is achieved without hampering their performance. The presented results are also shown to outperform conventional methods.

Index Terms—Internet of Things (IoT), Authentication System, Zero Knowledge Proof (ZKP), HMAC Algorithm, IoT Devices, Performance Metrics.

I. INTRODUCTION

The Internet of Things is now bringing communication and data-sharing capabilities to various devices, thus reshaping the digital world. With the increase in IoT networks, security-related issues, such as device authentication and secure communication, are also increasing. Most of the traditional authentication methods are not enough for the IoT network environment, because of its computational and memory hunger which becomes infeasible for resource-constrained IoT devices. The Zero Knowledge Proof protocol is a special cryptographic protocol that allows one party, known as a prover, to be able to prove to another party, called the verifier, that a given statement is true without leaking any information other than the statement's validity [1]. One of the well-known techniques used in ZKP is the Fiat-Shamir Protocol, which transforms interactive proof systems into non-interactive proof systems, making it perfect for the stateless nature of IoT entities [2].

In this system, devices can transmit, for example, rather than a password itself, proofs created from an HMAC hash of the selected password—"God is great" This way, even if the proof were to be intercepted by other people, it would be a very difficult task to reverse-engineer and reveal the original password.

The HMAC hashing algorithm combines a cryptographic key with a cryptographic hash function. Hence, most of the key calculations and metrics in assessing the performance of the proposed authentication system is done with respect to its efficiency in real-world IoT applications. Accuracy should authenticate legitimate devices properly while rejecting unauthorized ones, a key factor in security assurance and trust in IoT networks. Shorter response times are crucial indicators for efficient authentication in real-time IoT applications. Memory Consumption is extremely important for IoT devices, which often have quite limited memory resources; in this regard, the system's memory footprint should be minimal to guarantee broad implementation. Finally, the throughput measures the quantity of data successfully transmitted and received in the process of authentication; high throughput is an indication of a strong, efficient system [3]. Both these metrics together provide an overall view in terms of the performance of the system that it should live up to for secure IoT deployments.

This paper includes the following contributions:

1. This work presents an integration of ZKP and HMAC algorithms, which uses proofs to authenticate IoT devices in the simulated network.
2. Using the simulation, features like accuracy, throughput, latency, memory consumption, and response time acquired are evaluated for the IoT devices.

There are five sections in this study. A brief overview of IoT networks, Zero Knowledge Proof (Fiat-Shamir Protocol), and the HMAC function is given at the start of Section I. A review of the literature on IoT networks, Zero Knowledge Proofs, and HMAC-based algorithms employed in the prior work is discussed in the second section. The elements of the suggested methodology are described in the third section. The fourth section describes the experimentation setup within the simulation and authentication process overview of the authentication system. The performance parameters of the system are based on approaches to the main topic of the fifth section, which focuses on results.

II. LITERATURE SURVEY

The study [6], explained broadly about the Internet of Things, focusing on its history, key technologies, and the diverse applications it can be applied to. Of course, the most

important of these is that IoT fundamentally involves the combination of sensors, communication networks, and data processing capabilities. Also how the Internet of Things can be applied in various sectors, such as healthcare, smart cities, agriculture, and industrial automation is discussed. The paper further describes the challenges of IoT, like security, privacy, and standardization, which, according to the authors, require more research for these challenges to be met.

The paper [7] proposes the fast growth of IoT devices and the corresponding escalation of security risks, therefore creating a heightened need for strong measures in this area. The various layers of IoT architecture are sensitive to cyber vulnerabilities: device, network, and application layers. It further goes on to touch on various available security solutions with their limitations and points out gaps that are necessary to be filled to make up a holistic IoT security solution. The paper stresses the call for developing new security frameworks that will mitigate new threats in the IoT ecosystem.

The paper [5] presented that they have focused on the benefits of edge computing, and therefore, the reduction in latency, improvement in computation, better security, and real-time data processing. It has explained the efficiency that is realized through bringing data processing close to the source. They have highlighted security challenges that are faced by the IoT devices in edge computing networks, such as lightweight encryption, resource constraints, and efficiency preservation.

The study [8] has utilized the Fiat-Shamir protocol in comparison to the Guillou-Quisquater protocol and emphasized the secure secret transmission without disclosure, highlighting the use of 512-bit memory for secure authentication. Also discussed the limitations of smart cards and problems of zero-knowledge protocol in authentication.

The system [4], proposes Contiki OS-based Cooja simulator is used to implement a network environment suitable for the authentication system. They have experimented with various zero-knowledge and HMAC-based authentication schemes in the simulator to evaluate the execution time and memory consumption. The comparison of the efficiency and security of the Fiat-Shamir protocol with the Guillou-Quisquater scheme, noting the former's faster performance and lower memory requirements, albeit with reduced security. In this case, the Fiat-Shamir identification scheme outperforms Guillou-Quisquater's in terms of speed and HMAC-MD5 uses less memory and is faster than implemented HMAC-based authentication schemes. In this study, they have drawbacks of not evaluating parameters like accuracy, throughput, and scalability of authentication systems using those authentication schemes.

The system [9], proposes a modified zero-knowledge proof algorithm for IoT authentication. In their study, they compared their modified algorithm with the previous version of the Fiat-Shamir protocol used in the previous study. It has enhanced the performance of the algorithm through discrete logarithmic calculations and exponent factors added to the protocol. Much focus is directed toward data integrity; their approach employs AES encryption, showing high performance and being very

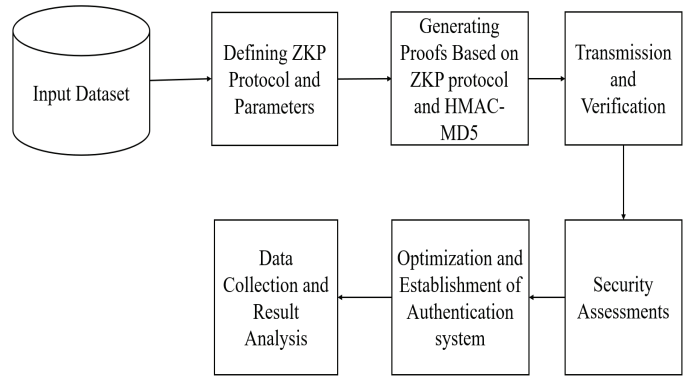


Fig. 1. System overview of proposed authentication system.

secure in comparison to earlier approaches. The shortcomings of this study include the failure to assess factors such as accuracy, memory usage, and scalability of the authentication system utilizing the authentication technique, which may have lower values.

The study [10] analysed the effectiveness of HMAC in securing cloud transactions. They have demonstrated HMAC implementation as a cloud transaction security mechanism with a very high success rate. For the experiment, they sent 1,000 requests of which, without applying HMAC, 620 requests were successful, while with HMAC, 870 requests were successful. The shortcomings of this study include, using HMAC in authentication systems can help to get more secure implementation in cloud networks.

The authors in the study [11] compared zero-knowledge proof with the conventional methods of authentication in web applications. They concentrated on data integrity, so they implemented zero-knowledge proof by using AES encryption. They proposed greater security in web application authentication compared to the conventional method, along with secure data transmission by the use of AES encryption implemented through the programming language Python. This study provides a fair discussion on the conventional method of login, its workflow of registration, authentication using the transmitting of hashed password, and the challenges of its security. However, it may have some weaknesses in that factors like accuracy, memory usage, and scalability of the authentication system using the authentication technique might have low-valued assessments.

Finally, the work in paper [12] reported on a novel library for the implementation of zero-knowledge proofs in low-resource computing devices. They show feasibility for zk-SNARKs on platforms such as Raspberry Pi, which shows promise for a new paradigm of secure computation on constrained devices.

III. THE PROPOSED METHODOLOGY

In the proposed methodology for the authentication system, the system overview is shown in Figure 1. System overview of the proposed authentication system. The overview includes input dataset, defining ZKP protocol and parameters, creating

proofs based on ZKP protocol and HMAC-MD5, transmission of proofs with verification, and security assessments, optimization of the authentication system and its establishment, and data collection along with result analysis.

In the simulated network used for the authentication system, in the initial phase generation of the input dataset is done which includes a large number of usernames and passwords to be used in the authentication process. It is essential to avoid using weak passwords as they can be vulnerable to default password attacks. The input of usernames and passwords are uniquely used by each server and client node.

In the next phase, defining of Zero Knowledge Proof (ZKP) protocol, specifically the Fiat-Shamir Protocol with HMAC-MD5 and their parameters is done. The authentication system leverages the Fiat-Shamir Protocol, in conjunction with the HMAC(MD5) algorithm, and generates proofs in the next phase of the system.

After that, transmissions and verifications of proofs over the network are implemented. After performing verification, the client node is authenticated by the server node and is either allowed or not allowed to form a secure connection along with sharing data.

In the security assessment phase, an equal number of attempts per node is set for equal load distribution. In the simulation environment, the authentication system undergoing the security test is being evaluated by increasing the number of attempts, which increases the stress on the system during the authentication process.

The next phase undergoes performing optimization in the coding part and algorithm. After this design ensures secure and efficient authentication without transmitting actual passwords, making it particularly suitable for resource-constrained IoT devices. The collection of simulation output data and result analysis is executed in the final phase.

The proposed authentication scheme uses some discrete logarithmic calculations to hike the system security without inflicting a heavy computational overhead on the nodes of the IoT. There are three primary phases in the system: Initialization, Registration, and Authentication.

An initialization is done for a group G and a random number g . At this stage, the user is to register; thus, they enter the username and password. The password is hashed using HMAC-MD5 to obtain x , and then $Y = g^x \mod n$ is computed. Then username and Y are sent to the server that stores them in its database as (username, Y, n).

In the Authentication phase, the server creates a one-time token a and random value c , then sends it to the client. The client rehashes his password and computes Y . Using a random value v , he computes $t = g^v \mod n$ and $r = v - cx$. The client transmit (username, r, t, a) to the server. The server shall now authenticate by checking whether $t = (Y^c \cdot g^r) \mod n$. If the authentication is successful, then the client is authenticated and can send data.

The key components include the group G , generator g , prime n , hashed password x , and various values that are specific to a session: a

IV. EXPERIMENTATION

A. Simulation and Evaluation

The system used for simulation is a machine with an Intel CPU having a Xeon 3.60 GHz processor, 2TB HDD, 32GB of RAM, and an Nvidia graphics card Quadro K620. To simulate the authentication system, the Cooja Simulator based on Contiki OS 3.0 is used. The system [4] uses the same network protocol with setting. The IEEE 802.15.4 communication standard (wireless) is adopted as a medium; channel modeling is real-world conditions including path loss, interference, and fading. The system used 10 nodes in the proposed system for five iterations of the simulation to accumulate parameters that have been evaluated for better results.

In the simulation environment, 10 nodes with one node being a server node, represent an typical IoT device. This way, the simulation model everything right from registration and proof generation to success in connection, data sharing, and disconnection. Such an all-inclusive approach is capable of allowing the analysis of system performance under working conditions that resemble real ones. To reach reasonably effective testing environments, this simulation is also factored in important aspects of Transmission Power and Node Mobility. For IoT devices, the power is set to a common value to give the appropriate communication range in an approximately real-world scenario. This setting helps evaluate the performance and reliability of the proposed system under realistic scenarios and measures how the performance of authentication varies with node mobility when the node is kept static throughout the entire simulation.

Since the nodes are stationary, this fixed configuration guarantees a focused examination of the processes for authentication in which the performance metrics correctly shows the potential of the system under a steady state. Initialization for further operations is initiated with the deployment and configuration of the IoT nodes and a server. In the Registration Phase, a client submits a registered username and password that falls within the dataset used by the server. It accepts access to knowledge within the set by the server node to realize the client with its username when the initial proof is received. A secure authentication framework that executes through the development and preservation of 128-bit HMAC(MD5) digests according to a password. Proofs are created and verified in the system to execute integrity in the authentication process and have been architected employing the Fiat-Shamir Protocol at particular intervals. To arrive at critical information, such as the number of nodes that have been authenticated, memory consumption, latency, response time, and throughput, such simulation performance indicators are tracked and collected in the Data Collection stage.

V. RESULTS

Important insights are revealed by the TABLE I. performance study of the IoT device authentication mechanism. In this, the parameters are evaluated across five iterations in the authentication process. The Parameter A is memory

consumption in bytes averaged at 1625.18 bytes. The average value for Parameter *A* remained at 218.08 bytes/second during the test, whereas Parameter *B* is nothing but the throughput in bytes/seconds. The Parameter *C* which is latency has an average value of 31.5 milliseconds. The Parameter *D* response time is averaged at 567.39 milliseconds with minimum fluctuations.

With the Parameter *E*, which has an accuracy rate of 98% on average, the system is often quite steady. The system has remained essentially stable despite the anomalies seen in the third iteration because it demonstrated the capacity to manage authentication jobs with high throughput and low latency.

TABLE I
PERFORMANCE OF THE LIGHTWEIGHT AUTHENTICATION SYSTEM FOR
IoT DEVICES

No of iterations	Parameter A (bytes)	Parameter B (Bytes/second)	Parameter C (milliseconds)	Parameter D (milliseconds)
1	1603.035	218.1	30.1	569.95
2	1614.715	218.5	31.3	568.69
3	1670.8776	216.8	34.5	569.45
4	1623.882	218.0	31.2	563.97
5	1613.373	219.0	30.4	564.91
Average	1625.17652	218.08	31.5	567.394

Comparing the simulation findings to conventional systems [11] tested in the same simulation environment as the lightweight system, the system's efficiency and performance are revealed, underscoring its potential for real-world Internet of Things applications. This project seeks to contribute to the creation of secure IoT networks, where devices can connect and share data with trust, by concentrating on lightweight and efficient authentication.

A. System Comparison

TABLE II showing A comparison between conventional and lightweight authentication systems reveals significant variations based on critical performance parameters.

TABLE II
COMPARISON OF CONVENTIONAL AND LIGHTWEIGHT SYSTEMS FOR
PERFORMANCE PARAMETERS

Parameters	Conventional	Lightweight
Parameter A (bytes)	1729.18026	1625.17652
Parameter B (bytes/second)	204.46	218.08
Parameter C (milliseconds)	589.568	567.394
Parameter D (milliseconds)	34.08	31.5
Parameter E (percentage)	96	98

In the Figure 2. Comparison of Conventional and Lightweight Systems for Parameter *A*. The lightweight system shows 6.01% a reduction which is 104 Bytes, indicating a more efficient use of resources. Also, in Figure 3. Comparison of Conventional and Lightweight Systems for Parameter *B*, observed improvement in throughput of the lightweight system by 6.66% which is 13.62 bytes/second, suggesting better handling of data or requests.

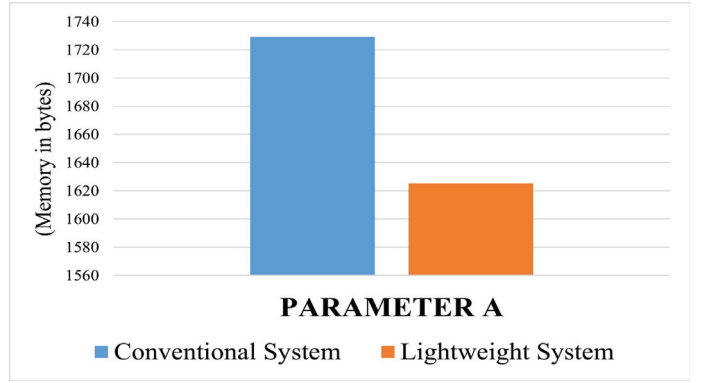


Fig. 2. Comparison of Conventional and Lightweight Systems for Parameter A.

In Figure 4. Comparison of Conventional and Lightweight Systems for Parameter *C* and Parameter *D*, the lightweight system performs better by 3.76% in terms of response time in comparison with the conventional system, which shows faster processing. In the case of latency also reduced in the lightweight system by 7.57% indicating quicker reaction times. Finally, the lightweight system achieves a higher accuracy rate by 2% compared with the conventional system shown in Figure 5. Comparison of Conventional and Lightweight Systems for Parameter *E*, demonstrating improved reliability and precision in authentication.

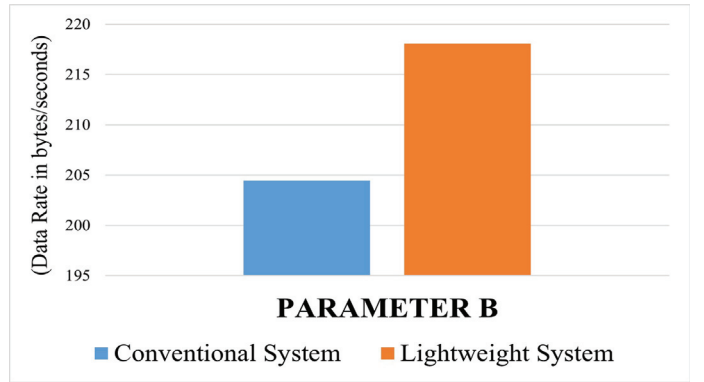


Fig. 3. Comparison of Conventional and Lightweight Systems for Parameter B.

All things considered, the suggested system performs better on all metrics—lower memory usage, higher throughput, quicker reaction times, less latency, and more accuracy—making it more dependable and lightweight for authentication needs.

Enhancements in network protocols and data processing algorithms can sustain high throughput over longer periods of time and higher data volumes. This system looks to handle data more efficiently than existing algorithms, potentially providing better performance in high-load situations. Data handling mechanisms can be optimized to smooth out any abrupt peaks and distribute the load more evenly.

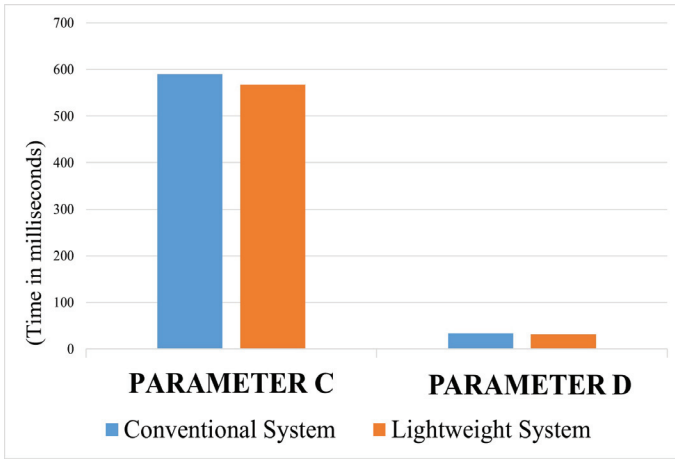


Fig. 4. Comparison of Conventional and Lightweight Systems for Parameter C and Parameter D.

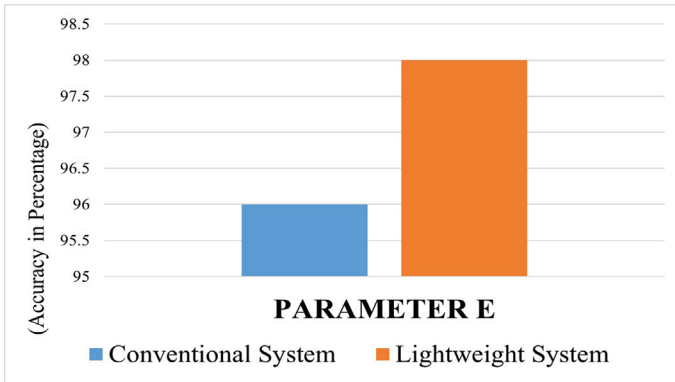


Fig. 5. Comparison of Conventional and Lightweight Systems for Parameter E.

VI. CONCLUSION

The lightweight authentication system's implementation of the Zero Knowledge Proof Protocol with the HMAC hashing algorithm presents a promising solution for IoT device authentication challenges. This paper presents a novel, lightweight, and reliable authentication system for IoT devices using the Zero Knowledge Proof Protocol and HMAC hashing algorithm. Our comprehensive evaluation via the Simulator reveals that the lightweight system excels in memory consumption by 6.01%, response time by 3.76%, latency by 7.57%, throughput by 6.66%, and accuracy by 2%, outperforming conventional methods. Its suitability for resource-constrained environments makes it a reliable solution for growing IoT networks.

Future research will concentrate on memory optimization, improving response time, throughput, scalability, accuracy, and practical implementation. Furthermore, investigating multi-factor authentication systems and sophisticated cryptography methods will improve security and guarantee the system's resistance to changing threats. By improving the system's performance and dependability, these improvements hope to encourage wider use in a range of IoT network scenarios.

REFERENCES

- [1] Will Major, William J. Buchanan, Jawad Ahmad, "An authentication protocol based on chaos and zero-knowledge proof" *Nonlinear Dyn* (2020) 99:3065–3087 Published online: 21 January 2020. <https://doi.org/10.1007/s11071-020-05463-3>
- [2] Crisitina Vilchez Moya, Juan Ramon Bermejo Higuera, Javier Bermejo Higuera, Juan Antonio Sicilia Monta Ivo, "Implementation and Security Test of Zero-Knowledge Protocols on SSI Blockchain", Accepted: 27 April 2023 / Published: 29 April 2023. <https://doi.org/10.3390/app13095552>
- [3] Phillip Williams, Indira Kaylan Dutta, Hisham Daoud, Magdy Bayoumi, "A survey on security in internet of things with a focus on the impact of emerging technologies" Volume 19, August 2022, 100564. <https://doi.org/10.1016/j.iot.2022.100564>
- [4] Leki Chom Thunon, Ravinandan, and Md. Iftexhar Hussain, "Performance Evaluation of Zero Knowledge and HMAC-Based Authentication in Fog-Based Internet of Things", Springer Nature Singapore Pte Ltd. 2021. <https://doi.org/10.1007/978-981-33-4084-861>
- [5] Abhay Verma and Vinay Verma, "Comparative Study of Cloud Computing and Edge Computing: Three Level Architecture Models and Security Challenges." *International Journal of Distributed and Cloud Computing*, June 2021. <http://www.publishingindia.com/ijdcc>
- [6] P.V. Dudhe, N.V. Kadam, R. M. Hushangabade, and M. S. Deshmukh, "Internet of Things (IOT): An Overview and its Applications", *International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS) 2017*, doi:10.1109/ICECDS.2017.8389935.
- [7] Sanjeev Kumar and Sukhvinder Singh Deora, "Security Challenges and Issues in IoT", 6th International Conference on Signal Processing, Computing and Control (ISPPC) 2021, doi:10.1109/ISPPC53510.2021.9609486.
- [8] Laleh Golafshan Mona Taghavi Ahmed Patel, Kenan Kalajdzic, "Design and Implementation of a Zero-Knowledge Authentication Framework for Java Card." *International Journal of Information Security and Privacy*, Volume 5, Issue 3, July 2011. <https://doi.org/10.4018/IJISP.2011070101>
- [9] Benfano Soewito and Yonathan Marcellinus, "IoT security system with modified Zero Knowledge Proof algorithm for authentication", *Egyptian Informatics Journal* Volume 22, Issue 3, September 2021, Pages 269-276. <https://doi.org/10.1016/j.eij.2020.10.001>
- [10] Dr. S. Justin Samuel and S. Judy Jenitha, "IEEE International Conference on Computational Intelligence and Computing Research", 2014. <https://doi.org/10.1109/ICCIC.2014.7238548>
- [11] Lum Jia Jun, Brandon, "Implementing Zero-Knowledge Authentication with Zero Knowledge (ZKA_wzk)", *The Python Papers Monograph 2*, 9 Proceedings PyCon Asia-Pacific, 2010.
- [12] Xavier Salleras and Vanesa Daza, "ZPiE: Zero-knowledge Proofs in Embedded systems," *Mathematics* 9, no. 20, page no. 2596, 2021.
- [13] Vaghani Divyeshkumar, "Block-chain based data provenance and integrity verification", revised on 10 June 2024; accepted on 13 June 2024. <https://doi.org/10.30574/ijscra.2024.12.1.1076>
- [14] Michał Goworko, Jacek Wyrębowicz, "A Secure Communication System for Constrained IoT Devices—Experiences and Recommendations" 2021 Oct; 21(20): 6906. Published online 2021 Oct 18. doi:10.3390/s21206906.
- [15] Mohammad Wazid, Ashok Kumar Das, Neeraj Kumar, Athanasios V. Vasilakos, "Design of secure key management and user authentication scheme for fog computing services", Volume 91, February 2019, Pages 475-492. <https://doi.org/10.1016/j.future.2018.09.017>